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Magnon-Cherenkov effect from a picosecond strain pulse

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Abstract

Radiation from an uniformly moving source, referred to as the Cherenkov effect, is a universal phenomenon, which enables emission of waves and finds important applications in various areas of physics from particle physics to plasmonics and beyond. Currently, unveiling the potential of Cherenkov emission of coherent spin waves, or magnons, is hampered by a lack of experimentally implemented fast-moving magnetic perturbations. In this work, we uncover the magnon-Cherenkov effect demonstrating experimentally the emission of the exchange spin waves enabled by an optically induced picosecond strain pulse, which acts as a spatially localized propagating perturbation of the internal effective magnetic field as a result of magnetoelastic coupling. Through time-resolved measurements and micromagnetoelastic simulations, we reveal the sagittal propagation of a strain pulse in a dielectric ferrimagnet followed by the emitted SWs fully in line with the conditions of the Cherenkov effect. The spectral characteristics of the emitted spin waves are controlled with an applied magnetic field and the shape of the strain pulse itself. Establishing a picosecond strain pulse as a source for the magnon-Cherenkov effect significantly expands the possibilities to realize and control non-dissipative spin transport in various laterally and vertically structured magnonic devices.

Keywords: Cherenkov effect, spin waves, magnonics, picosecond acoustics

Magnonic devices use coherent spin waves (SWs) to encode, transfer, and process information [1]. Considerable advancement has been made in recent years in the development of planar two-dimensional (2-D) magnonic structures, featuring GHz magnetostatic waves that are guided laterally from their area of origin [2]. As a result, magnonic elements with diverse functionalities are actively being developed for analog and digital computing [3] with a potential to outperform conventional electronic-based schemes in terms of scaling down, operating frequencies and power consumption [4, 5]. In addition to the implementation of classical binary logic operations [6, 7], non-conventional computation paradigms computation paradigms [8], including magnonic neuromorphic computing [9] are emerging. Scaling down, bursting operation frequencies, and increasing the density of magnonic elements demand means to generate SWs

beyond the magnetostatic regime, *i.e.* possessing short wavelengths and high velocities, with low damping, and propagating in any lateral or sagittal direction in magnetic heterostructures [10]. Currently, the two main paradigms to excite coherent SWs exploit either localized alternating magnetic field or spin current, or propagating acoustic waves coupled to SWs. However, as waves of any nature, coherent SWs can be generated via the Cherenkov mechanism, which is the fundamental process of wave emission from a uniformly moving and spatially localized source [11]. The breakthrough discovery of electromagnetic wave emission from uniformly moving charged particles by Pavel Cherenkov [12] was awarded the Nobel Prize in 1958, and the Cherenkov effect has been investigated in multiple fields of physics and beyond: particle physics [13], plasmonics [14] and biomedicine [15]. The potential of this mechanism for magnonics lies in the accessibility of the exchange SWs and the possibility of guiding them by controlling the direction of propagation of the source, which requires the search for a suitable moving perturbation of the magnetic media [16].

In analogy to the Cherenkov effect for electromagnetic and sound waves, such an emission of SWs requires the perturbation of a magnetic medium which moves with a velocity matching the SW phase velocity V_{ph} and is sufficiently localized in the direction of propagation. In a one-dimensional scenario, an infinitely localized magnetic field pulse of strength H_0 exemplifies the concept of Cherenkov emission of SWs. This pulse propagates through a magnetically ordered medium along the z axis at the velocity V, and is represented by the Dirac delta function $H(t,z) = H_0\delta(z - Vt)$, where t is time (Fig. 1a). The impact of the field on magnetic moments is described by the torque $T(z,t) = T_0\delta(z - tV)$, where T_0 is its magnitude. This spatially localized torque is known to trigger SWs propagating in the $\pm z$ direction with the distribution of wavenumbers k defined by the Fourier transform (FT) of the torque $T(z,t)$ [17–19]. Thus, the Dirac delta torque enables the generation of SWs with any k when considering that $T(z)$ is fixed in space, and the corresponding frequencies ω are given by the SW dispersion relation $\omega(k)$. When the pulse moves through a media, 2-D FT of the torque $\hat{\delta}(kV + \omega)$ yields that only SWs satisfying the velocity matching (VM) condition will be generated:

$$V = |\frac{\omega}{k}| \equiv V_{\mathrm{ph}}(k). \tag{1}$$

The torque of a finite spatial size and an arbitrary shape imposes a constraint on the amplitudes of the SWs depending on their wavenumbers [20]. When this torque is moving, SWs emitted due to the Cherenkov effect are determined by the VM-condition with additional restriction given by its spectrum $\hat{T}(k)$:

$$\hat{T}(\omega, k) = \hat{T}(k)\hat{\delta}(kV + \omega). \tag{2}$$

Thus, to realize Cherenkov emission of SWs, a source producing a spatially localized torque with a required spectrum and moving with the SW phase velocity is required.

In Ref. [21], an experimental realization of the magnon-Cherenkov effect is demonstrated with magnetic-field flux quanta produced by moving Abrikosov vortices in a superconductor/ferromagnet structure, which serve as moving localized magnetic field pulses. This experiment confirms the excitation of monochromatic exchange SWs with wavelengths as short as 35 nm at cryogenic temperatures. Nevertheless, the problem remains of an implementation of a fast-moving localized field at room temperature with the possibility to propagate in various directions and controllable spectral characteristics. A perspective candidate for a propagating torque is a moving domain wall [22, 23], which acts locally on a magnetization, and an extremely narrow wall can be considered in the same way as the moving Dirac pulse with the VM-condition defined by equation (1). However, the Walker breakdown limits the propagation velocity of domain walls in flat structures, and an experimental demonstration is challenging and yet to be realized. Here, magnetic nanotubes are a promising platform to overcome the speed limit [24]. Laser pulses are known to produce effective magnetic field pulses, *i.e.* a perturbation of internal fields, via various femtomagnetic processes [25, 26]. However, with the large group and phase velocities, such pulses satisfy the VM-condition only at very low k, resulting mostly in the excitation of the uniform precession of magnetization.

In this work, we exploit the propagating effective field provided by a bipolar picosecond strain pulse, which can be implemented at room temperature in various media [27]. Consider a mechanical displacement of atoms $\mathbf{u_z}(z,t)$ in the form of a Gaussian function moving with a sound velocity, which induces an effective field through a perturbation of the magnetoelastic energy and the associated torque $T(z)$ (Fig. 1a). In general, the dispersion curves of the exchange SWs and acoustic phonons have two intersection points, marked with the brown points in Fig. 1b,c, where the VM-condition (1) is satisfied, and the Cherenkov emission of the SWs is expected to occur. In addition, the amplitude of the emitted SWs is given by the Fourier amplitude at a particular $k_{\mathrm{I,II}}$, which is controlled by the spatial length of the displacement pulse (Fig, 1b). The significant difference in the group velocities of SWs at the intersection points of the dispersion curves results in the propagation of the SWs with k_{I} and k_{II} behind and ahead of the strain pulse, respectively, as schematically illustrated in Fig. 1a. In the following, we experimentally uncover the magnon-Cherenkov effect resulting in the emission of the exchange SWs from the picosecond strain pulse propagating sagittally in iron garnet and support the experimental results with micromagnetoelastic simulations.

Observation of Cherenkov emission of spin waves

We explore the Cherenkov emission of SWs on the platform of the 1.74-μm-thick film of lutetium iron garnet $(LuBi)_3(FeGa)_5O_{12}$ (LuIG) grown on the gadolinium gallium garnet (GGG) substrate. The sample was partially covered with a 100-nm-thick gold layer (Au) that serves as an optoacoustic transducer [28]. Linearly polarized femtosecond laser pump pulses with the central wavelength of 515 nm are focused either onto the Au layer to generate a sagittally propagating strain pulse via a thermoelastic mechanism or onto the uncovered LuIG film to directly excite magnetization dynamics (see Fig. 2a and Methods). The time-resolved response is detected at the opposite interface of the LuIG film as a change of intensity or polarization rotation of the 450 nm probe pulses reflected from the LuIG/GGG interface. The central wavelengths of the pump and probe pulses guarantee their spatial separation in LuIG, since the penetration depths of both pulses are below 100 nm.

To verify that the strain pulse is indeed generated in the structure, we first measure pump-induced changes in the reflectivity. The resulting time trace consists of high-frequency oscillations with increasing amplitude, which emerge slightly after the pump pulse impact on the Au/LuIG (Fig. 2b). Then, the oscillations experience an abrupt change at $t \approx 0.26$ ns, when the frequency is reduced and the envelope acquires a complex shape (see inset of Fig. 2b). The fast Fourier transform (FFT) spectrum of the entire time trace comprises three main frequency peaks (dark blue bottom spectrum in Fig. 2c): the low-frequency group of peaks around 13 GHz, the narrow peak at 55.2 GHz and the broad peak at 69.4 GHz. The sagittal propagation of the acoustic pulse results in a time-domain Brillouin scattering (TDBS) and corresponding oscillations in the measured time traces. The mechanism behind the TDBS oscillations lies in the interference of the probe pulse reflected from the LuIG/GGG interface with that reflected from the propagating strain pulse [29]. We observe that the time trace before $t = 0.26$ ns reveals the only TDBS frequency 69.4 GHz (gray middle spectrum in Fig. 2c) corresponding to the strain pulse propagating in the LuIG with the velocity 6.55 km s^{-1} of the longitudinal acoustic (LA) phonons [30]. The rest of the time trace consists of the oscillations with the TDBS frequency 55.2 GHz originating from the strain pulse in GGG with the LA sound velocity of 6.2 km s^{-1} along with the low-frequency group of peaks (light blue top spectrum in Fig. 2c). Therefore, the strain pulse is produced on the Au/LuIG side of the sample and propagates to the LuIG/GGG interface. To identify the dynamics corresponding to the low-frequency peaks, we study the transient magneto-optical response.

In the probe polarization rotation measurements, when the pump pulse is focused onto the Au/LuIG, the magneto-optical response is identified as oscillations with an amplitude and frequency dependent on the external magnetic field. Such oscillations occur at time delays exceeding $\Delta t \approx 0.26$ ns, *i.e.* the magnetization dynamics follows the strain pulse (blue line in Fig. 2d and Extended Data Fig. E1). In contrast, direct excitation of the uncovered LuIG film yields the excitation of the magnetization dynamics right after the pump impact (red line in Fig. 2d). The FFT of these time traces reveals that the two main

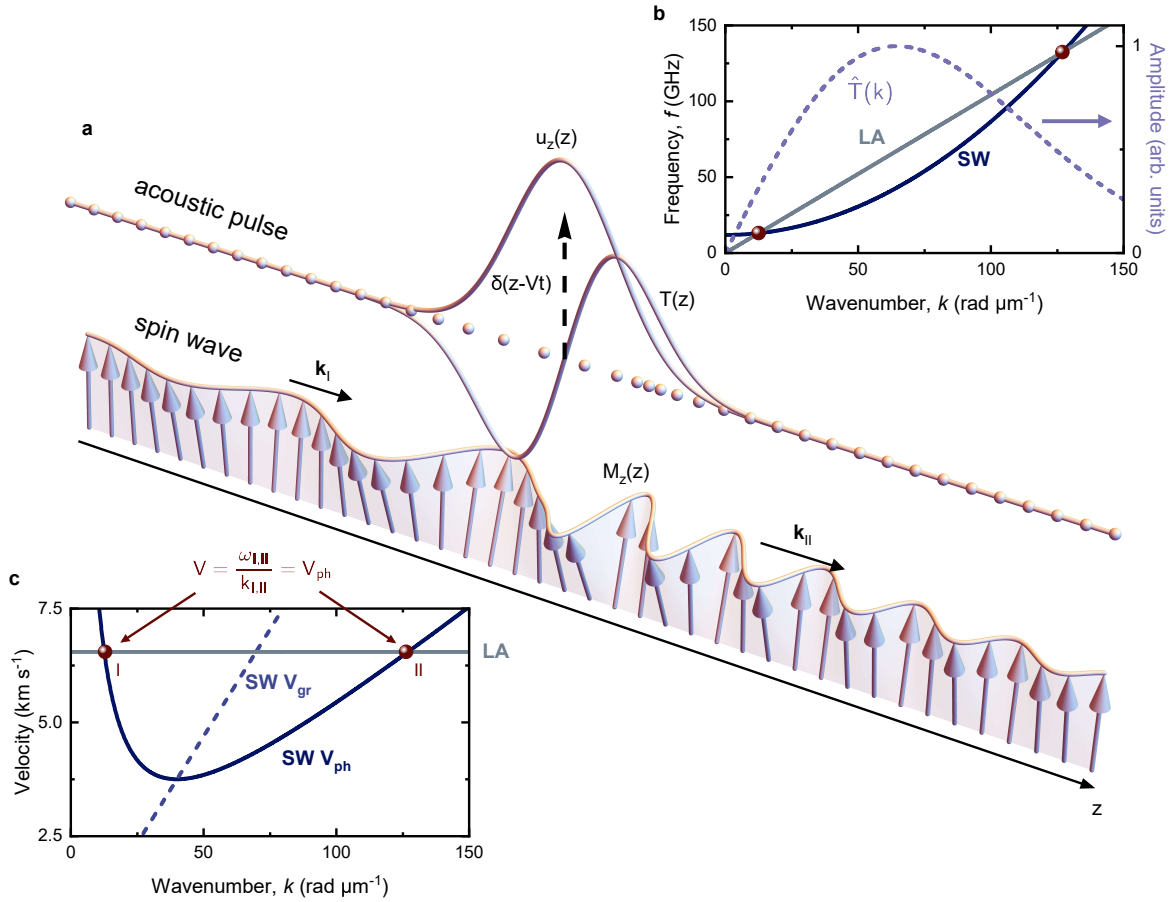


Fig. 1 The Cherenkov emission of spin waves from a propagating strain pulse. a, The longitudinal mechanical displacement $u_z(z)$ with a Gaussian profile induces a localized effective magnetic field and associated torque $T(z)$ acting on magnetization. The VM-condition is imposed by the convolution of $T(z)$ and moving Dirac delta function (black dashed arrow), resulting in the excitation of SWs with wavenumbers k_{I} and k_{II}. **b**, The dispersion curves of the exchange SWs (dark blue) and the LA phonon (gray) with two intersection points corresponding to the VM-condition(1). The strain pulse of a particular spatial length implies the exact amplitude spectrum $\hat{T}(k)$ shown by purple dashed line. **c**, SW phase velocity V_{ph} (dark blue), SW group velocity V_{gr} (dashed purple) and the LA phonon (gray) is plotted against the wavenumber. The VM-condition is satisfied as the velocity of the LA phonons exceeds the minimal SW V_{ph}. The corresponding points are marked with the brown circles.

frequency peaks (f_0 and f_{I}) have different amplitude ratios when the pump is focused onto the Au/LuIG and directly onto the LuIG (Fig. 2e). We attribute the lowest frequency peak (11.9 GHz) to a sagittally quasi-homogeneous precession of magnetization at the ferromagnetic resonance (FMR) frequency, which dominates under the direct LuIG excitation. The excitation of the FMR mode is routinely realized in experiments with picosecond acoustic pulses [29, 31], and is observed under the Au/LuIG excitation in our experiment as well. The higher frequency peak (13 GHz) is more pronounced when the strain pulse is generated in the Au/LuIG structure, corresponds to the SW with non-zero k, and can be a signature of the magnon-Cherenkov effect.

To confirm the Cherenkov emission of SWs, we study the magnetization dynamics as a function of the strength of the external magnetic field H_{ext}. The FFT of the corresponding time traces at $\mu_0 H_{\mathrm{ext}}$ between 0.3 and 0.7 T reveals that f_0 and f_{I} depend on the field in a distinct manner (Fig. 3a). The FMR frequency f_0 shows linear dependence on the field. Note that the same trend of f_0 is observed under the direct excitation of the LuIG (inset in Fig. 3a). The frequency f_{I} grows non-linearly with H_{ext}, signifying

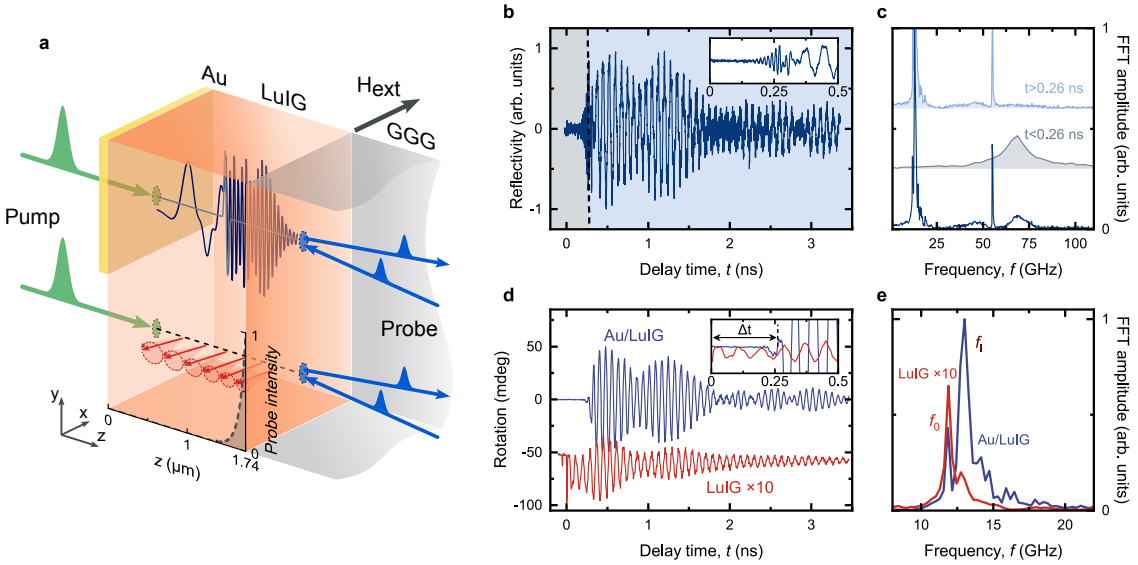


Fig. 2 Time-resolved optical excitation of the strain pulse and detection of SWs. a, The scheme of the experiments when the pump pulses are focused either onto the Au/LuIG or onto the LuIG. Probe pulse is focused onto the LuIG/GGG interface, and its in-depth intensity profile is shown by gray area. **b**, The time trace of the reflectivity changes after the pump pulse impact. Gray and light blue regions before and after $t = 0.26\,\mathrm{ns}$ mark the propagation of the strain pulse in the LuIG film and in the GGG substrate, respectively, Inset shows the zoom of time window before 0.5 ns. **c**, The FFT amplitude spectra obtained from the data on **b**: bottom (dark blue) spectrum corresponds to the FFT of the whole time trace, the middle (gray) and top (light blue) spectra correspond to the FFT with sharp window edge before and after $t = 0.26\,\mathrm{ns}$, respectively. Spectra are shifted vertically for clarity. **d**, The time traces of the probe polarization rotation angle after the impact of the pump pulse focused onto the Au/LuIG (blue) and the LuIG (red, scaled by factor of 10 and shifted vertically by -50 mdeg for clarity). Inset shows the time shift between the emergence of the oscillations $\Delta t \approx 0.26\,\mathrm{ns}$. **e**, The FFT amplitude spectra obtained from the data on **d**. The external magnetic field is $\mu_0 H_{\mathrm{ext}} = 0.4\,\mathrm{T}$.

that this is the frequency of the emitted SW, which satisfies the VM-condition for the magnon-Cherenkov effect.

As the VM-condition (1) imposes the specific ratio $\frac{\omega}{k} = V_{\mathrm{s}}$, where V_{s} is the LA phonon velocity in our case, we find the emitted SW frequency as a function of H_{ext}. The exchange SWs dispersion relation is

$$\omega(H_{\mathrm{ext}}, k) = \omega_0(H_{\mathrm{ext}}) + Dk^2, \tag{3}$$

where $\omega = 2\pi f$ is the SW angular frequency, $\omega_0 = 2\pi f_0$ is the FMR angular frequency, $D = 2A_{\mathrm{ex}}\gamma M_s^{-1}$ is the spin stiffness, A_{ex} is the exchange constant, $\gamma = 1.76 \times 10^{11}\,\mathrm{rad\,s^{-1}\,T^{-1}}$ is the gyromagnetic ratio, and M_s is the saturation magnetization. Solving the LA phonon dispersion relation $\omega(k) = V_{\mathrm{s}}k$ and equation (3) together, we obtain the frequency of SWs emitted via the magnon-Cherenkov effect:

$$\omega_{\mathrm{I,II}} = \frac{V_{\mathrm{s}}}{2D}\left(V_{\mathrm{s}} \mp \sqrt{V_{\mathrm{s}}^2 - 4D\omega_0}\right), \tag{4}$$

where the minus and plus signs correspond to the SWs at frequencies $\omega_{\mathrm{I}} = 2\pi f_{\mathrm{I}}$ and $\omega_{\mathrm{II}} = 2\pi f_{\mathrm{II}}$, respectively.

We acquire the LA phonon dispersion with $V_{\mathrm{s}} = 6.55\,\mathrm{km\,s^{-1}}$ and the exchange SW dispersion using equation (3) with $A_{\mathrm{ex}} = 1.8 \times 10^{-12}\,\mathrm{J\,m^{-1}}$ in the range of the experimentally used H_{ext} (Fig. 3b). As H_{ext} increases and the exchange SW dispersion shifts towards higher frequencies, the wavenumber of the emitted SW k_{I} (k_{II}) increases (decreases), as shown by the solid (dashed) brown line in Fig. 3b. The resulting field dependencies of the Cherenkov radiation frequencies f_{I} and f_{II} are non-linearly approaching

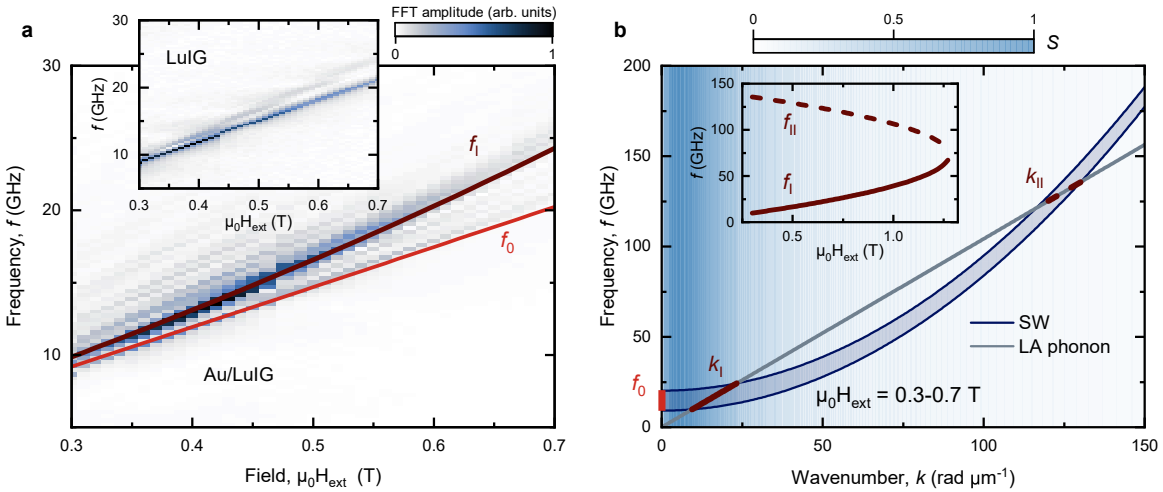


Fig. 3 Magnetic field dependence of the excited magnetization dynamics. a, The field dependence of the FFT spectra acquired in time-resolved magneto-optical measurements when the pump is focused onto the Au/LuIG (color-coded). Solid lines correspond to the FMR frequency f_0 (red) and to the frequency of the Cherenkov radiation f_I (brown). Inset in **a** demonstrates the field dependence when the pump is focused directly onto the LuIG. **b**, Calculated SW dispersion (blue) in the range of the external magnetic field $\mu_0 H_\mathrm{ext}$ between 0.3 and 0.7 T and the LA phonon dispersion (gray). The VM-condition of the Cherenkov radiation is satisfied in the two intersection points of the dispersion curves, which shift with H_ext is traced by the brown solid (I) and dashed (II) lines. With the increase of H_ext the wavenumber k_I and frequency f_I of the emitted SWs increase, while k_II and f_II decrease according to the shift of the exchange SW dispersion curve (inset in **b**). Color code of the background represents the sensitivity $S(k)$ of the 450 nm probe laser pulses in reciprocal space.

each other (inset in Fig. 3b). The field dependence of f_I demonstrates good agreement with that obtained experimentally (brown line in Fig. 3a). It follows that the observed SW at f_I is the wave emitted from the strain pulse via the magnon-Cherenkov effect and lagging behind its source. The exchange SWs frequencies f_I observed in the experiments correspond to wavelengths ranging from 0.27 to 0.5 μm. SWs with frequency f_II and wavelength between 49 and 54 nm are not detected in the experiment.

Micromagnetoelastic simulations

We support our interpretation of the experimental results as the strain pulse-driven Magnon-Cherenkov effect with micromagnetoelastic simulations. We numerically solved the system of coupled Landau-Lifshitz-Gilbert equation for magnetization $\mathbf{M}(\mathbf{r}, t)$ and the elastodynamic equation (Navier-Cauchy) for mechanical displacement $\mathbf{u}(\mathbf{r}, t)$. The strain pulse is given a Gaussian profile $u_z(z, t)$ with a total duration of 20 ps and injected into LuIG, and the magnetization dynamics in response to the pulse propagation is calculated. Details of the numerical methods are presented in Ref. [32] and Methods.

A snapshot of the spatial distribution of the out-of-plane magnetization component over the sagittal direction $M_z(z)$, the displacement $u_z(z)$ and the strain tensor component $\varepsilon_{zz}(z)$ at $t = 141$ ps are shown in Fig. 4a and d. It is seen that there are long-(short-)wavelength SW propagates behind (ahead) the strain pulse. These SWs are continuously emitted as the strain pulse traverses the film (see Supplementary Movie 1). The probe pulse with penetration depth $\delta_\mathrm{pr} = 75$ nm is sensitive to a small portion of the magnetization dynamics averaged over a narrow region near the LuIG/GGG interface, thus limiting the wavenumber resolution (see gray area in Fig. 4a and Methods). Therefore, SWs with wavelengths shorter than 186 nm (k larger than 33.7 rad μm^{-1}) cannot be resolved in the experiments, justifying why SWs with frequency f_II are not present in the experimental data. We employ spatial averaging with exponentially decaying weights to obtain $M_z(t)$ time trace to account for the penetration depth available in the experiment.

Turning to Fig. 4b, we observe that the magnetization dynamics emerges exactly at $\Delta t = 0.26$ ns after the injection of the strain pulse. Its FFT spectrum consists of the large peak corresponding to the exchange

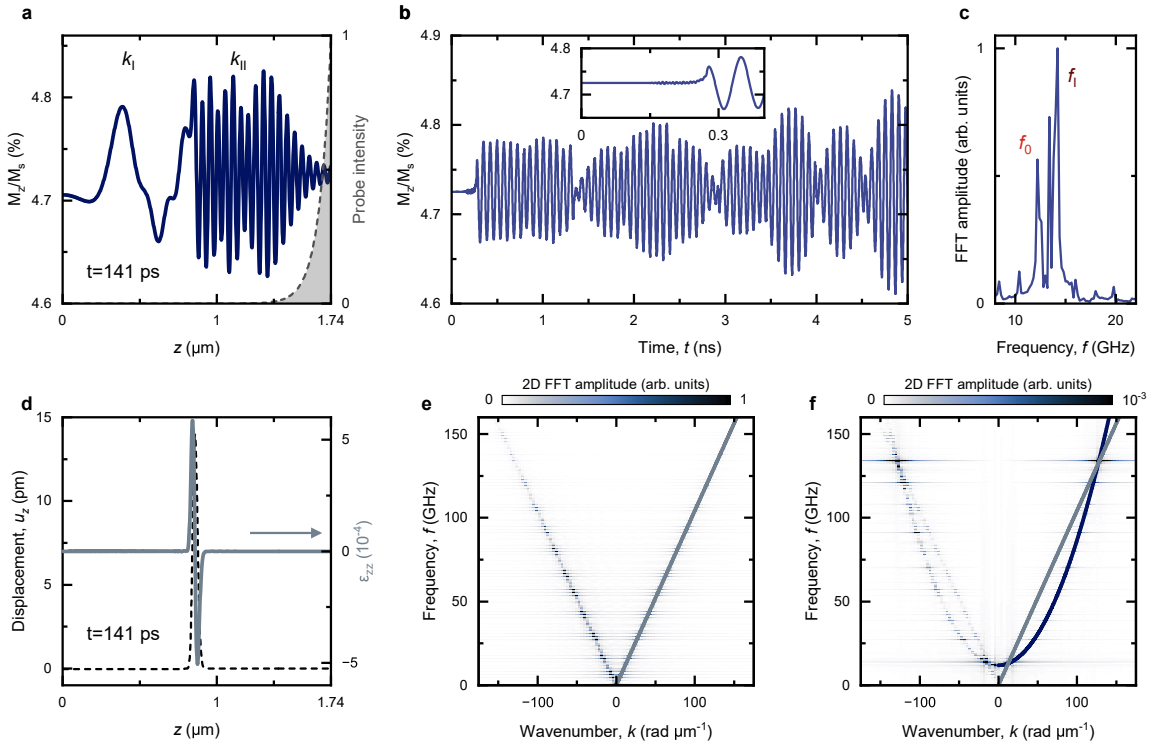


Fig. 4 Micromagnetoelastic simulations of the Cherenkov emission of the SWs from the strain pulse a, Spatial snapshot of the normalized out-of-plane M_z component at $t = 141\,\mathrm{ps}$ after the strain pulse injection. Gray area corresponds to the intensity profile of 450 nm probe with penetration depth $\delta_{\mathrm{pr}} = 75\,\mathrm{nm}$. **b**, Time trace $M_z(t)$ obtained by spatial averaging with weights corresponding to the probe penetration profile. **c**, The FFT spectrum of the time trace from **b**. **d**, Spatial snapshot of the mechanical displacement u_z (dashed line) and $\frac{\partial u_z}{\partial z}$ corresponding to the strain $\varepsilon_{zz}(z)$ (solid line) at $t = 141\,\mathrm{ps}$. Strain pulse duration is 20 ps. **e**, Dispersion characteristic of the simulated acoustic pulse, obtained by the 2-D FFT of the spatio-temporal evolution of u_z. Gray solid line corresponds to the LA phonon dispersion with $V_s = 6.55\,\mathrm{km\,s^{-1}}$. **f**, Dispersion characteristic of the simulated magnetization dynamics obtained by the 2-D FFT of the spatio-temporal evolution of $M_z(t, z)$. Gray line corresponds to the LA phonon dispersion, and dark blue curve corresponds to the exchange SWs dispersion calculated with equation (3). The external magnetic field is $\mu_0 H_{\mathrm{ext}} = 0.4\,\mathrm{T}$.

SW with frequency f_{I} and of the less pronounced FMR peak with frequency f_0, which agrees with the experimentally obtained values (compare Fig. 4c and Fig. 2e). To detect the SW with wavenumber k_{II} one needs to reduce δ_{pr} by tuning the probe wavelength into the ultraviolet range (Extended Data Fig. E3 and Methods). Finally, the dispersion characteristic of the magnetization dynamics is obtained using the 2-D FFT of the simulated spatio-temporal evolution of $M_z(z, t)$ and compared to analytical ones (Fig. 4f). The increased amplitude of SWs at the intersections of the dispersion curves serves as confirmation of the magnon-Cherenkov effect responsible for the SW emission.

Discussion

The group velocities of the emitted SWs are in the range between 1.2 and $2.2\,\mathrm{km\,s^{-1}}$ for k_{I} and between 10.9 and $11.9\,\mathrm{km\,s^{-1}}$ for k_{II}, which differ significantly from the strain pulse velocity (Fig. 1c) and lead to SWs propagation behind and ahead of the strain pulse (Fig. 4a,d). However, SW with k_{II}, propagating ahead of the strain, is not observed in the experiments. Although the VM-condition is satisfied for both intersection points, equation (2) yields the amplitude condition on the magnon-Cherenkov effect, which is $\hat{T}(k) = \exp\left[-\frac{1}{2}k^2\sigma^2\right]|k|\sigma$ for the Gaussian displacement pulse with spatial length σ. Assuming that SWs with a particular k are emitted with sufficient amplitude only if $\hat{T}(k)$ exceeds 10% of its maximum value,

we obtain that the SW wavenumber should be in the range between $0.06\sigma^{-1}$ and $2.76\sigma^{-1}$. For the 20 ps long strain pulse ($\sigma = 16$ nm), this range is from 4 to 170 rad μm^{-1}. Evidently, for longer strain pulses, the range of accessible SW wavenumbers reduces and shifts toward the lower k. We numerically confirm this showing the absence of SW with k_{II} and the increased amplitude of SW with k_{I} during the Cherenkov emission from the strain pulse with a duration of 70 ps (Extended Data Fig. E2 and Supplementary movie 2). This is fully consistent with conservation laws of momentum and energy for the magnon-phonon system, which also implies a modification of the initial spectrum of the strain pulse by emitted SWs. Consequently, the strain pulse acquires a small distortion during propagation. At the same time, the 2-D FFT of the spatio-temporal evolution of $u_z(z,t)$ shows that the phonon dispersion remains linear (see Fig. 4e), indicating an absence of a strong hybridization between the emitted SWs and the strain pulse.

Since the Cherenkov emission accompanies the strain pulse during its propagation, we attempted to detect these SWs using probe pulses with 900 nm central wavelength. Since the studied LuIG sample is almost transparent at 900 nm, such probe pulses are sensitive to the magnetization dynamics averaged over the entire sample thickness. When the pump pulse is focused directly onto the LuIG, only the magnetization dynamics at the FMR frequency f_0 is detected. Under the excitation of the Au/LuIG the magnetization dynamics is detected without a time delay (Extended Data Fig. E4). The FFT spectra of this dynamics show, in addition to the FMR mode, the presence of spin waves generated through the Cherenkov effect. Moreover, distinct frequency combs appear in the spectra, arising from the strain pulse and corresponding SWs reflecting multiple times between the sample interfaces.

In the outlook, we discuss the possible impact of the demonstrated magnon-Cherenkov effect on the development of magnonics. Unlike other propagating SW sources, such as domain walls, vortices, and even light pulses, strain pulses generally have favorable propagation properties in various media and heterostructures. The magnon-Cherenkov effect therefore enables generation of SWs from strain pulses moving within magnetic/magnetic or magnetic/nonmagnetic multilayers, as well as in patterned structures [31, 33, 34]. Here, SWs in different constituents remain coherent but exhibit distinct characteristics determined by the specific magnetic and acoustic properties of the layers involved. Thus, complex magnetic dynamics, controlled by external magnetic fields, could find applications in reservoir computing [35]. On the other hand, one could envisage multilayer or patterned structures with a large acoustic impedance mismatch at a particular interface, thus making it possible to inject SW without injection of the strain pulse through the interface, or even through an air spacer [36]. This makes it possible to physically decouple the source of an exchange SW and a waveguide. By designing optoacoustic transducers, sequences of strain pulses with controlled distance can be obtained and utilized for a unidirectional Cherenkov emission of SW in analogy to the numerical concept proposed in [37]. Regarding the development of approaches to realize Cherenkov emission of a SW, we note that in heterostructures, such as magnonic crystals [38] and in hybrid magnon-phonon crystals [39], the velocity threshold can be modified in an analogy to Cherenkov electromagnetic radiation by a charged particle moving in a photonic crystal [40]. An intriguing possibility to generate SW by acoustic solitons [27] has yet to be explored to access exceptionally high SW wavenumbers owing to the strong localization of mechanical displacements to be utilized as, *e.g.* sources of THz electromagnetic waves [41]. Finally, as the SWs excitation is possible through dynamical perturbations of exchange, spin-orbital, and magneto-dipolar effective fields, mechanisms of SW generation from a strain pulse beyond magnetoelastic coupling can be explored.

Methods

Sample

The sample is a single-crystal film $(LuBi)_3(FeGa)_5O_{12}$ with a thickness of 1.74 μm, grown by a liquid-phase epitaxy method on (110) gadolinium gallium garnet (GGG) substrate. The surface of the sample was covered with a 100-nm-thick gold layer using a thermal evaporation process. The 256×256 μm^2 windows in gold are made using electron beam lithography combined with reactive ion etching by an Ar-ion plasma.

The magnetic parameters of the sample were characterized using the Smit-Beljers approach [42] in combination with the pump-probe FMR frequency measurements. We performed FFT of time traces at different azimuthal directions of the magnetic field with $\mu_0 H_{ext} = 415\,\mathrm{mT}$ (Extended Data Fig. E5), when the pump pulses are focused onto the Au/LuIG and the probe pulses central wavelength of 1030 nm. The free energy density for substituted iron garnets grown on the (110)-type substrates is written in a form [43]:

$$U = E_H + E_M + E_u + E_i + E_{xz}, \tag{5}$$

where E_H and E_M are Zeeman and magnetostatic energy terms, respectively, as well as E_u, E_i and E_{xz} are responsible for the uniaxial out-of-plane, in-plane and orthorhombic anisotropy, respectively. All terms are expressed as follows:

$$\begin{aligned} E_H &= -\mu_0 M_s (m_x H_x + m_y H_y), \\ E_M &= \frac{1}{2}\mu_0 M_s^2 m_z^2, \\ E_u &= -K_u m_z^2, \\ E_i &= -K_i m_x^2, \\ E_{xz} &= -K_{xz} m_x m_z, \end{aligned} \tag{6}$$

where $m_j = M_j/M_s$ $(j = x, y, z)$ are the normalized components of the magnetization $\mathbf{M}$; H_x and H_y are the in-plane components of the external magnetic field $\mathbf{H}_{\mathrm{ext}}$; K_u, K_i and K_{xz} are the uniaxial out-of-plane, in-plane and orthorhombic anisotropy parameters, respectively. Writing (5) in the spherical coordinates allows a numerical calculation of the FMR frequency, knowing the equilibrium direction of $\mathbf{M}$, as described elsewhere [42, 44]. By fitting the azimuthal and field dependence of FMR frequency (solid red line in Extended Data Fig. E5 and red dashed line in Fig. 3a) we acquire the following magnetic parameters of the sample: $M_s = 13.5\,\mathrm{kA\,m^{-1}}$, $K_u = 194\,\mathrm{J\,m^{-3}}$, $K_i = 449\,\mathrm{J\,m^{-3}}$ and $K_{xz} = 325\,\mathrm{J\,m^{-3}}$.

Pump-probe experiment details

We used the Yb:KGW regenerative amplifier laser system (PHAROS, Light Conversion) as a source of laser pulses with a duration of 170 fs, central wavelength of 1030 nm, 5 kHz repetition rate, and pulse energy up to 1 mJ. We split the laser pulses into two parts. The first part was used as a linearly polarized pump pulse with a central wavelength of 515 nm after frequency doubling with β-barium borate (BBO) crystal. The second part of the initial pulses was used to pump the optical parametric amplifier (ORPHEUS, Light Conversion), which produced linearly polarized pulses with a duration of 150 fs and a central wavelength of 900 nm. We used the second BBO crystal to achieve the probe pulses with a central wavelength of 450 nm.

The pump pulses after passing a mechanical delay line were focused onto the sample in a spot with a diameter of 50 μm with typical fluence of $50\,\mathrm{mJ\,cm^{-2}}$. The probe pulses were focused from the opposite side through an optically transparent GGG substrate in a spot with a diameter of 25 μm and a fluence of at least 10 times less than for the pump pulses. An angle of incidence of probe pulses was 17°, and a spatial filter was used on the reflected beams. The scheme allows distinguishing the probe pulses reflected from the different sample interfaces and, therefore, to exclude parasitic back-reflection from the substrate. Pump-induced rotation of the probe's polarization was measured using the balanced photodetector (New Focus 2107, Newport) in combination with the Wollaston prism and half-wave plate. The signal from the detector was demodulated by the lock-in amplifier (SR830, Stanford Research Systems) synchronized with the optical chopper placed in the pump pulse optical path. The sample was placed inside the electromagnet to apply the external magnetic field in the sample's plane. All experiments were carried out at room temperature.

Probe laser pulses sensitivity

The probe pulse of 150 fs duration sets an upper frequency limitation of an order of THz and does not obscure the detection of SWs of interest. Detection of the propagating SW through the Bragg mechanism used in Ref.[45] requires the matching between the SW and the light wavevectors and is not applicable in our case. Indeed, the probe pulse of 450 nm is sensitive to the SW with $k = 66\,\mathrm{rad}\,\mu\mathrm{m}^{-1}$, and SW frequency lies between $f = 44.5\,\mathrm{GHz}$ and $f = 52.8\,\mathrm{GHz}$ at used H_{ext}, which is not observed in our experiments. However, the strongly decaying 450 nm probe intensity profile in the sample itself enables the detection of sagittally propagating magnetization disturbances. With the FT of the exponential depth profile of the probe intensity, we obtain the spectral amplitude S of SW with particular k in the detected signal, which scales with the probe penetration depth δ_{pr} as:

$$S(k) = \frac{1}{\sqrt{1 + (k\delta_{\mathrm{pr}})^2}}. \tag{7}$$

Thus, the use of the 450 nm probe pulses with $\delta_{\mathrm{pr}} = 75\,\mathrm{nm}$ enables efficient detection of SWs with k up to $33.7\,\mathrm{rad}\,\mu\mathrm{m}^{-1}$ determined as $S(k) = e^{-1}$ in equation (7), which corresponds to the wavelength of 186 nm (value $S(k)$ is color-coded background in Fig.3b). The frequency of SW detected in the experiments f_{I}, according to equation (3) corresponds to the wavenumbers k_{I} in the range from 12.6 to $23.3\,\mathrm{rad}\,\mu\mathrm{m}^{-1}$. SW with these wavenumbers appear to be detected efficiently, while f_{II} SW with k_{II} from 116 to $127\,\mathrm{rad}\,\mu\mathrm{m}^{-1}$ are not resolved.

Micromagnetoelastic simulations details

The interrelated dynamics of magnetization $\mathbf{m}(\mathbf{r},t) = \mathbf{M}(\mathbf{r},t)/M_{\mathrm{s}}$ and mechanical displacement $\mathbf{u}(\mathbf{r},t)$ were described by a numerical solution of a system of Landau-Lifshitz-Gilbert (LLG) and Navier-Cauchy equations accounting for the two-way magnetoelastic coupling between them. The LLG equation was solved in the form

$$\frac{d\mathbf{m}}{dt} = -\gamma\mathbf{m}\times\mathbf{H}_{\mathrm{eff}} - \alpha\gamma\mathbf{m}\times(\mathbf{m}\times\mathbf{H}_{\mathrm{eff}})\,, \tag{8}$$

where γ is the (absolute value of) electron's gyromagnetic ratio and α is the Gilbert damping parameter. The effective magnetic field $\mathbf{H}_{\mathrm{eff}}$ acting on magnetization was calculated via $\mathbf{H}_{\mathrm{eff}} = -1/(\mu_0 M_s)\frac{\partial F}{\partial \mathbf{m}}$, where F is the free energy density of LuIG including the magnetoelastic term in the form

$$\begin{aligned} F_{\mathrm{mel}} = B_1(\varepsilon_{xx}m_x^2 + \varepsilon_{yy}m_y^2 + \varepsilon_{zz}m_z^2) + \\ + B_2\left[(\varepsilon_{xy}+\varepsilon_{yx})m_xm_y + (\varepsilon_{xz}+\varepsilon_{zx})m_xm_z + (\varepsilon_{yz}+\varepsilon_{zy})m_ym_z\right], \end{aligned} \tag{9}$$

where ε_{ij} are the elastic strains, $B_1 = 3.39\times10^6\,\mathrm{erg\,cm^{-3}}$ [46] and $B_2 = 5.5\times10^6\,\mathrm{erg\,cm^{-3}}$ [47] are the magnetoelastic constants of LuIG assumed to be equal to those of YIG.

The Navier-Cauchy equation was solved in the form

$$\rho\frac{\partial^2\mathbf{u}}{\partial t^2} + \beta\frac{\partial\mathbf{u}}{\partial t} = \nabla\boldsymbol{\sigma}, \tag{10}$$

where ρ is the mass density, β is the elastic damping coefficient and $\boldsymbol{\sigma}$ is the Cauchy stress tensor including the magnetoelastic term $\boldsymbol{\sigma}_{\mathrm{mel}}$ having the components $\sigma_{ii}^{\mathrm{mel}} = B_1m_i^2$ and $\sigma_{ij}^{\mathrm{mel}} = \sigma_{ji}^{\mathrm{mel}} = B_2m_im_j$. We used $\rho = 7.39\,\mathrm{g\,cm^{-3}}$[30] and elastic moduli $c_{11} = 3.1705\times10^{12}\,\mathrm{erg\,cm^{-3}}$, $c_{44} = 0.6920\times10^{12}\,\mathrm{erg\,cm^{-3}}$.

The material was quantized using a computational cell size $\Delta z = 2$ nm in a 1-D geometry. It was assumed that $\mathbf{m}(z,t)$ and $\mathbf{u}(z,t)$ are homogeneous in the xy plane and can only change along the z axis corresponding to the LuIG out-of-plane dimension. However, the demagnetizing field was accurately calculated, taking into account the variation of $\mathbf{m}$ along z and an infinite-plane geometry in lateral dimensions [48]. LLG equation was solved using a 4-order projective Runge-Kutta method with time step

3×10^{-14} s, and Navier-Cauchy equation was solved using a finite difference approximation with the time step 1.5×10^{-14} s. The free magnetic boundary condition $\partial \mathbf{m}/\partial \mathbf{n} = 0$ was imposed on both surfaces of LuIG, where $\mathbf{n}$ is the normal vector to the surface, and the free-surface mechanical boundary condition $\sigma_{zi} = 0$ was employed at the surface $z = 0$, while the opposite surface at $z = 1.74\,\mu$m was described by a fixed boundary condition $\mathbf{u} = 0$.

In order to correctly describe the magnetoelastic dynamics, first we found an equilibrium distribution of $\mathbf{m}(\mathbf{H})$ and $\mathbf{u}(\mathbf{H})$ in a relaxed film of LuIG in a magnetic field $\mathbf{H}$. Then, the strain pulse was applied to the film in the relaxed state. Modeling of the pulse injection was performed by imparting a time-varied displacement $\mathbf{u}(z = 0, t)$ of such a form as to produce a bipolar Gaussian strain ε_{zz} with amplitude 5×10^{-4} and duration $\tau = 20$ or 70 ps, corresponding to the spatial lengths σ of 16 and 23 nm. In the calculations, a Gilbert damping value $\alpha = 10^{-3}$ was used. We assigned the elastic damping parameter β an arbitrarily small value of $\beta/\rho = 2 \times 10^8\,\mathrm{s}^{-1}$.

Data availability

All the experimental data from this work are available through figshare at https://doi.org/10.6084/m9.figshare.28351496.

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Declarations

The authors declare no competing interests.

Extended Data

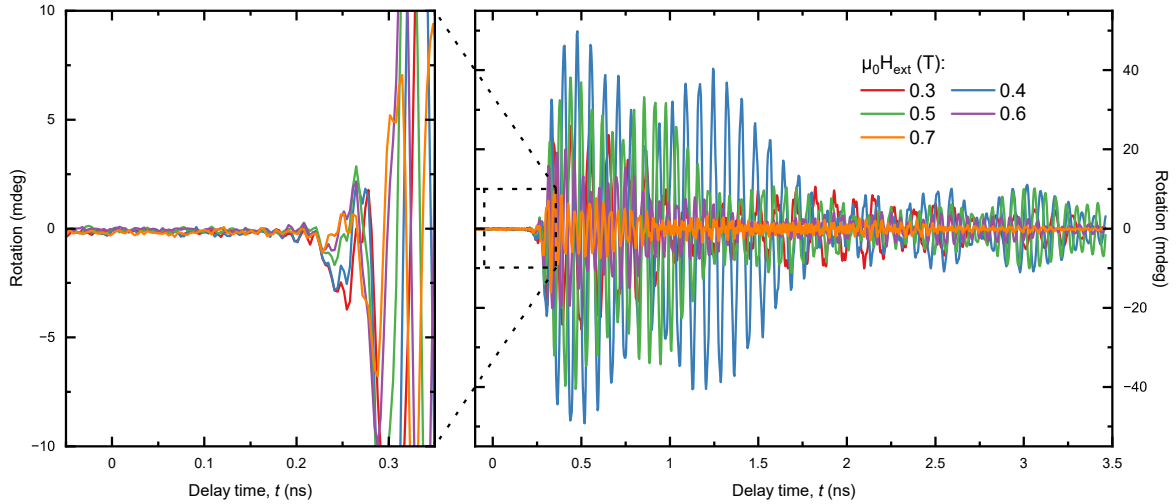


Fig. E1 Experimental time traces of 450 nm probe pulses polarization rotation under different H_{ext} obtained when the pump is focused onto the Au/LuIG. The left panel shows the enlarged area of the right one (dashed box).

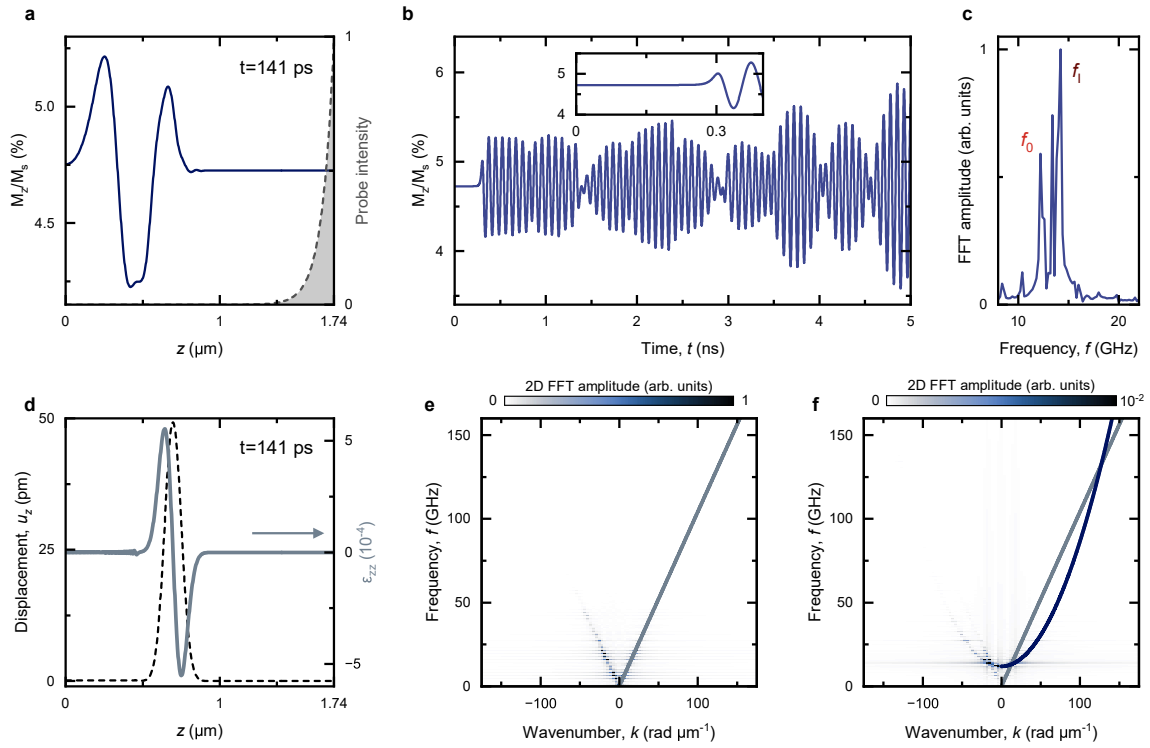


Fig. E2 Micromagnetoelastic simulations of the Cherenkov emission of the SWs by the strain pulse with duration of 70 ps a, Spatial snapshot of the normalized out-of-plane M_z component at $t = 141$ ps after the strain pulse injection. Gray area corresponds to the intensity profile of 450 nm probe with penetration depth $\delta_{\mathrm{pr}} = 75$ nm. **b**, Time trace $M_z(t)$ obtained by spatial averaging with weights corresponding to the probe penetration profile. **c**, The FFT spectrum of the time trace from **b**. **d**, Spatial snapshot of the mechanical displacement u_z (dashed line) and $\frac{\partial u_z}{\partial z}$ corresponding to the strain $\varepsilon_{zz}(z)$ (solid line) at $t = 141$ ps. Strain pulse duration is 20 ps. **e**, Dispersion characteristic of the simulated acoustic pulse, obtained by the 2-D FFT of the spatio-temporal evolution of u_z. Gray solid line corresponds to the LA phonon dispersion with $V_s = 6.55\,\mathrm{km\,s^{-1}}$. **f**, Dispersion characteristic of the simulated magnetization dynamics obtained by the 2-D FFT of the spatio-temporal evolution of $M_z(t, z)$. Gray line corresponds to the LA phonon dispersion, and dark blue curve corresponds to the exchange SWs dispersion calculated with equation (3). The external magnetic field is $\mu_0 H_{\mathrm{ext}} = 0.4$ T.

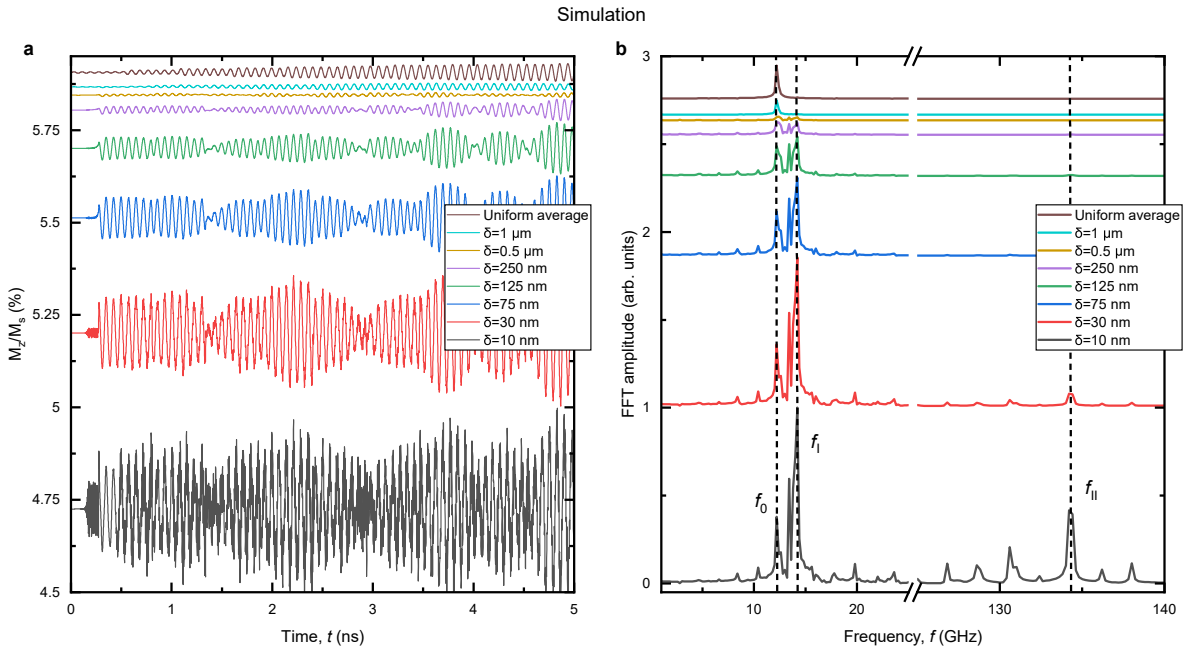


Fig. E3 Probe pulses sensitivity depending on the penetration depth. a, Simulated time traces of the out-of-plane magnetization component $M_z(t)$ obtained by spatial averaging with weights corresponding to the penetration profiles with different $\delta_{\rm pr}$. **b**, The FFT of the data from **a**.

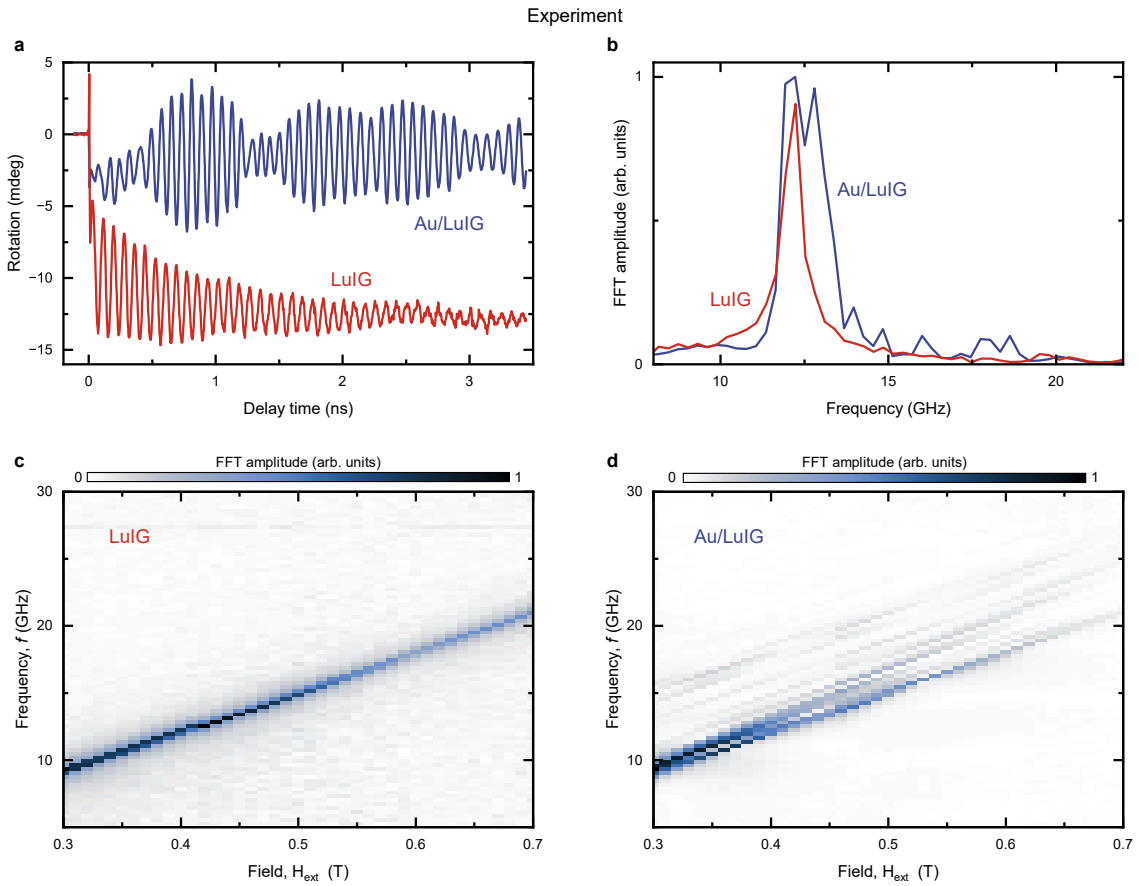


Fig. E4 Pump-probe experimental results obtained with probe pulses with central wavelength of 900 nm. **a**, Time traces of probe polarization rotation when the pump is focused onto the Au/LuIG (blue line) and onto the LuIG (red line). **b**, The FFT of the data from **a**. **c**,**d**, Field dependencies of the FFT of the polarization rotation time traces when the pump is focused onto the LuIG and onto the Au/LuIG, respectively.

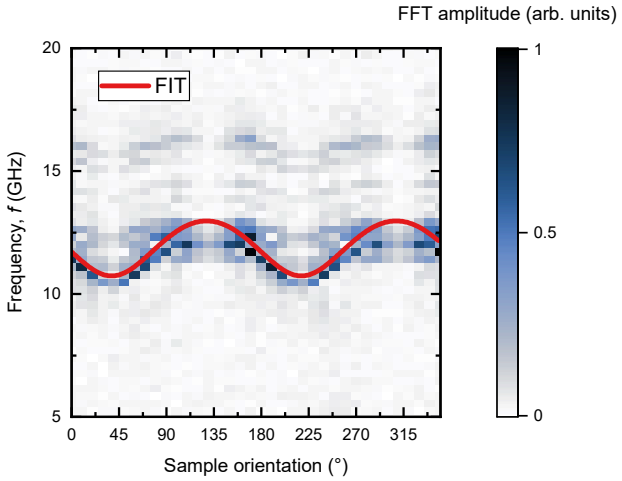


Fig. E5 Dependence of the magnetization dynamics frequency on the azimuthal orientation of the external magnetic field. The color map corresponds to the FFT of the time traces measured with the applied in-plane field $\mu_0 H_{\mathrm{ext}} = 415\,\mathrm{mT}$. The red line corresponds to the fit of the FMR frequency versus the in-plane orientation of the magnetic field, obtained using the Smit-Beljers approach.

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